

I will suggest realistic starting parameters for the CORRECTED model, explaining the choice of parameters based on physical systems, deriving them from typical experimental values for Rashba 2DEGs (like surface states of Au(111), Bi/Ag interfaces, or oxide interfaces like LaAlO₃/SrTiO₃).

I will ensure the units are consistent (using SI where possible, or standard condensed matter units like eV, Å, meV) and that the formulas for the corrected model (with $\hbar^2$ in denominator) are used to derive realistic Magnetization values.

Suggested Realistic Starting Parameters for the CORRECTED Model

1. Physical System Selection

To provide realistic starting parameters, we select a **Rashba Two-Dimensional Electron Gas (2DEG)** formed at the interface of an oxide heterostructure, specifically **LaAlO₃/SrTiO₃ (LAO/STO)**. This system is a canonical experimental realization of the Rashba effect with large spin-orbit coupling and tunable carrier density [Caviglia et al., *Phys. Rev. Lett.* 104, 166802 (2010)].

Alternatively, parameters for the **Bi/Ag(111) surface alloy** (a giant Rashba system) can be used, but the oxide interface is often more representative for transport-based Edelstein effect studies (current-induced spin polarization). We will provide parameters for a "generic" strong Rashba system that fits the LAO/STO or similar interface description.

2. Base Parameters (SI Units & Condensed Matter Units)

The following parameters define the physical properties of the electrons in the 2D plane.

2.1 Effective Mass (m)

- **Value:** $m = 0.5 m_e$ to $2.0 m_e$
- **Specific Suggestion:** $m = 1.0 m_e = 9.11 \times 10^{-31}$ kg.
- **Source/Justification:**
 - In LAO/STO, the effective mass is anisotropic and ranges from $0.5m_e$ to $3m_e$ depending on the sub-band occupancy [Caprara et al., *Rep. Prog. Phys.* 79, 096501 (2016)].
 - For a generic robust simulation, m_e is a standard starting point. In condensed matter units ($\hbar = 1$), mass is often expressed in $\text{eV}^{-1}\text{\AA}^{-2}$. Using $m = 1.0m_e$, we have:

$$m = \frac{1.0 \times 9.11 \times 10^{-31} \text{ kg}}{(1.05 \times 10^{-34} \text{ J} \cdot \text{s})^2} \approx 8.2 \times 10^{38} \text{ J}^{-1} \text{ m}^{-2}$$

Converting to eV and Å (1 J = 6.24 × 10¹⁸ eV, 1 m = 10¹⁰ Å):

$$m \approx \frac{1}{7.62 \text{ eV} \cdot \text{Å}^2} \approx 0.131 \text{ eV}^{-1} \text{ Å}^{-2}$$

- **Code Parameter:** `m_eff = 0.131` (in eV⁻¹ Å⁻²).

2.2 Rashba Spin-Orbit Coupling (α)

- **Value:** α = 10⁻¹¹ eV·m to 10⁻¹⁰ eV·m.
- **Specific Suggestion:** α = 5.0 × 10⁻¹¹ eV·m (50 meV·Å).
- **Source/Justification:**
 - Surface states of Au(111) have α ≈ 0.3 – 0.5 eV·Å.
 - LAO/STO interfaces typically have α ≈ 10 – 60 meV·Å depending on the electric field and density [Ben Shalom et al., *Phys. Rev. Lett.* 105, 206801 (2010)].
 - α = 50 meV·Å represents a **strong Rashba regime** where effects are easily observable.
 - In Si: α = 50 × 10⁻³ eV × 10⁻¹⁰ m = 5 × 10⁻¹² eV · m.
 - **Code Parameter:** `alpha = 0.05` (in eV*Å).

2.3 Fermi Energy (E_F) / Chemical Potential (μ)

- **Value:** 10 meV to 200 meV.
- **Specific Suggestion:** E_F = 60 meV (High Density Regime).
- **Source/Justification:**
 - The band crossing point is typically set to 0.
 - E_F is tunable via gating in experiments.
 - E_F = 60 meV places the system firmly in the **High Density Regime (HDR)** where both bands are occupied. The "Rashba energy" is E_R = $\frac{m\alpha^2}{2\hbar^2}$.
 - Calculating E_R:
Using m ≈ 0.13 eV⁻¹ Å⁻² and α = 0.05 eV · Å:

$$E_R = \frac{1}{2} m \alpha^2 = 0.5 \times 0.131 \times (0.05)^2 \approx 0.00016 \text{ eV} = 0.16 \text{ meV}$$

Wait, let's check units. m in eV⁻¹ Å⁻², α in eV Å.

$$m\alpha^2 \sim (\text{eV}^{-1} \text{Å}^{-2})(\text{eV}^2 \text{Å}^2) = \text{eV}.$$

$$E_R \approx 0.16 \text{ meV}.$$

- This seems very small. Let's re-evaluate m . In LAO/STO m^* can be heavy.
- If E_R is small, $E_F = 60 \text{ meV}$ is definitely in the HDR ($E_F \gg E_R$).
- To see interesting non-monotonic behavior, one might set E_F closer to E_R , but for "realistic" robust transport, E_F is usually significantly higher. Let's stick with $E_F = 60 \text{ meV}$.
- **Code Parameter:** $E_F = 0.06$ (in eV).

2.4 Transport Relaxation Time (τ)

- **Value:** 0.1 ps to 10 ps.
- **Specific Suggestion:** $\tau = 1.0 \times 10^{-12} \text{ s}$ (1 ps).
- **Source/Justification:**
 - Mobility $\mu = e\tau/m$. In LAO/STO, mobilities can range from 1 to 1000 $\text{cm}^2/\text{V}\cdot\text{s}$.
 - If $\mu \sim 10 \text{ cm}^2/\text{V}\cdot\text{s}$ and $m \sim m_e$, then $\tau = \mu m/e \approx \frac{0.1 \cdot 10^{-4} \cdot 10^{-30}}{10^{-19}} \approx 10^{-13} \text{ s}$.
 - High mobility samples reach $\tau \sim 10^{-11} \text{ s}$.
 - $\tau = 1 \text{ ps}$ is a realistic median value for "good quality" interfaces.
 - **Code Parameter:** $\tau = 1\text{e-}12$ (in seconds).
 - Note: If using eV units for time (where $1 \text{ s} \approx 6.6 \times 10^{15} \text{ eV}^{-1}$), $\tau \approx 6600 \text{ eV}^{-1}$.

2.5 Electric Field (E)

- **Value:** 10^2 V/m to 10^5 V/m .
- **Specific Suggestion:** $E_x = 1000 \text{ V/m}$ (1 kV/m).
- **Source/Justification:**
 - The Edelstein effect is a linear response effect. $M \propto E$.
 - Experimental setups for detecting spin accumulation typically apply currents corresponding to moderate fields ($\sim 10^3 \text{ V/m}$) to avoid heating/breakdown while maintaining a measurable signal.
 - **Code Parameter:** $E_x = 1000$ (in V/m).

2.6 Physical Constants

- Bohr Magnetron $\mu_B = 9.27 \times 10^{-24} \text{ J/T}$ (or $5.79 \times 10^{-5} \text{ eV/T}$).
- Elementary charge $e = 1.60 \times 10^{-19} \text{ C}$.
- Reduced Planck constant $\hbar = 1.05 \times 10^{-34} \text{ J}\cdot\text{s}$ (or $6.58 \times 10^{-16} \text{ eV}\cdot\text{s}$).

3. Corrected Model Formulas with Parameters

Using the dimensional correction identified in the analysis (adding $\hbar^{-2}$), we can predict the expected Magnetization order of magnitude for these parameters.

3.1 High Density Regime (HDR) Magnetization

Formula:

$$M_y = \frac{\mu_B |e| \tau}{2\pi \hbar^2} m \alpha E_x$$

Calculation:

1. Prefactor $A = \frac{\mu_B e \tau}{2\pi \hbar^2}$.

$$\hbar^2 \approx 1.1 \times 10^{-68} \text{ J}^2 \text{ s}^2.$$

$$e\tau \approx 1.6 \times 10^{-19} \times 10^{-12} = 1.6 \times 10^{-31} \text{ C} \cdot \text{s}.$$

$$\mu_B \approx 9.3 \times 10^{-24} \text{ J/T}.$$

$$A \approx \frac{10^{-23} \cdot 10^{-31}}{10^{-68}} \approx \frac{10^{-54}}{10^{-68}} \approx 10^{14} \text{ A} \cdot \text{m}^{-1} \text{T}^{-1} \text{V}^{-1} \text{s}^{-1} \dots$$

$$\text{Let's simplify units: } [A] = \frac{(\text{J/T})(\text{C})(\text{s})}{\text{J}^2 \text{s}^2} = \frac{\text{C}}{\text{T} \cdot \text{J} \cdot \text{s}} = \frac{\text{A}}{\text{T} \cdot \text{J} \cdot \text{s}^2} \dots$$

Actually, let's convert to Condensed Matter units for estimation.

$$\hbar = 6.58 \times 10^{-16} \text{ eV} \cdot \text{s}.$$

$$\hbar^2 \approx 4.3 \times 10^{-31} \text{ eV}^2 \text{s}^2.$$

$$m = 0.131 \text{ eV}^{-1} \text{A}^{-2}.$$

$$\alpha = 0.05 \text{ eV} \cdot \text{A}.$$

$$\tau = 1 \text{ ps} = 10^{-12} \text{ s}.$$

$$e = 1.6 \times 10^{-19} \text{ C}.$$

$$E_x = 1000 \text{ V/m}.$$

$$\mu_B = 5.79 \times 10^{-5} \text{ eV/T}.$$

$$M_y = \frac{1}{2\pi} \cdot \frac{(5.79 \times 10^{-5})(1.6 \times 10^{-19})(10^{-12})}{(6.58 \times 10^{-16})^2} \cdot (0.131) \cdot (0.05) \cdot (1000)$$

$$\text{Numerator Part 1 (Constants): } 9.2 \times 10^{-36}.$$

$$\text{Denominator Part 1: } 4.3 \times 10^{-31}.$$

$$\text{Ratio: } 2.1 \times 10^{-5}.$$

$$\text{Units of Ratio: } \frac{\text{eV} \cdot \text{C} \cdot \text{s}}{\text{eV}^2 \cdot \text{s}^2} = \frac{\text{C}}{\text{eV} \cdot \text{s}}.$$

Now multiply by $m\alpha E$:

$$m\alpha E = (0.131 \text{ eV}^{-1} \text{A}^{-2})(0.05 \text{ eV} \cdot \text{A})(1000 \text{ V/m}).$$

$$m\alpha E = 6.55 \text{ A}^{-1} \text{V/m}.$$

$$\text{Convert V to eV: } 1 \text{ V} \approx 6.24 \times 10^{18} \text{ eV}.$$

$$E = 1000 \text{ V/m} = 6.24 \times 10^{21} \text{ eV/m}.$$

$$m\alpha E = 6.55 \text{ A}^{-1} \cdot 6.24 \times 10^{21} \text{ eV/m} \approx 4 \times 10^{22} \text{ eV}/(\text{A} \cdot \text{m}).$$

Note: $\text{A} \cdot \text{m} = 10^{10} \text{ \AA} \cdot 10^{-10} \text{ m} \dots$ units are getting messy.

Simpler Approach:

The formula M_y should result in units of μ_B / Area (spin density).

Let's check the value of the susceptibility $\chi = M_y/E_x$.

$$\chi \approx \frac{\mu_B e \tau}{2\pi \hbar^2} m \alpha.$$

Plug in numbers:

$$\chi \approx \frac{(10^{-4} \text{ eV/T})(10^{-19} \text{ C})(10^{-12} \text{ s})}{10^{-30} \text{ eV}^2 \text{ s}^2} (0.13)(0.05).$$

$$\chi \approx \frac{10^{-35}}{10^{-30}} (6 \times 10^{-3}) \approx 10^{-5} \times 6 \times 10^{-3} = 6 \times 10^{-8}.$$

$$\text{Units: } \frac{\text{eV} \cdot \text{C} \cdot \text{s}}{\text{eV}^2 \cdot \text{s}^2} \cdot \text{eV}^{-1} \text{A}^{-2} \cdot \text{eV} \cdot \text{A} = \frac{\text{C}}{\text{eV} \cdot \text{s}} \cdot \text{A}^{-1} = \frac{\text{A} \cdot \text{s}}{\text{eV} \cdot \text{s} \cdot \text{A}} = \text{eV}^{-1}.$$

This implies $\chi \sim 10^{-8} \text{ eV}^{-1}$?

$$\text{If } M = \chi E, [M] = [\chi][E] = \text{eV}^{-1} \cdot \text{V/m} = \text{V}/(\text{eV} \cdot \text{m}) = 1/\text{m}.$$

This is not Magnetization (μ_B/m^2).

There is still a unit mismatch in the manual check because μ_B in eV/T converts magnetization to energy, not to spin density directly in these simplified units without accounting for the definition of E in eV.

Let's trust the SI derivation of the correction.

The order of magnitude for the Edelstein effect is typically characterized by the efficiency parameter λ_{IEE} (Inverse Edelstein Effect length) or equivalent.

$$\lambda_{IEE} \sim \alpha \tau / \hbar.$$

With $\alpha \sim 50 \text{ meV} \cdot \text{A}$, $\tau \sim 1 \text{ ps}$, $\hbar \sim 0.66 \text{ eV} \cdot \text{fs}$:

$$\alpha \tau \sim 50 \cdot 10^{-3} \text{ eV} \cdot 10^{-10} \text{ m} \cdot 10^{-12} \text{ s} = 5 \cdot 10^{-24} \text{ J} \cdot \text{m}.$$

$$\hbar \sim 10^{-34}.$$

Ratio $\sim 10^{10} \text{ m}$? No.

α (SI) $\approx 5 \times 10^{-12} \text{ J} \cdot \text{m}$? No, α is velocity in SI ($\sim 10^5 \text{ m/s}$).

$$\alpha_{SI} \approx 10^5 \text{ m/s}.$$

$$\lambda \sim \frac{10^5 \cdot 10^{-12}}{10^{-34}} \sim 10^{17} \text{ m}.$$

This suggests the formula $\lambda = \alpha \tau / \hbar$ is dimensionally wrong or α needs to be $\text{eV} \cdot \text{m}$.

If α is $\text{eV} \cdot \text{m}$ ($\approx 10^{-11}$), then $\alpha \tau \approx 10^{-23}$. Dividing by 10^{-34} gives 10^{11} m . Still huge.

The standard formula for induced spin density $S \sim e E \tau \alpha \nu$.

Density of states $\nu \sim m / \pi \hbar^2$.

$$S \sim e E \tau \alpha \frac{m}{\hbar^2}.$$

$$E \sim 10^3. \tau \sim 10^{-12}. \alpha \sim 10^{-11} (\text{eV m}). m \sim 10^{-30}. \hbar^2 \sim 10^{-68}.$$

$$S \sim 10^{-19} \cdot 10^3 \cdot 10^{-12} \cdot 10^{-11} \cdot \frac{10^{-30}}{10^{-68}}.$$

$$S \sim 10^{-38} \cdot 10^{38} \sim 1.$$

So $S \sim 1 \mu_B / \text{m}^2$? No, units of e is C.

Result $\sim$ Coulomb / meter squared $\sim$ Current density / velocity?

Spin density is dimensionless ($\hbar/2$) or $1/\text{m}^2$.

The calculation 10^0 suggests $S \sim 1$ unit of "something".

Realistic spin densities are on the order of $10^{10} - 10^{12} \mu_B / \text{cm}^2$ (which is $10^{14} - 10^{16} \mu_B / \text{m}^2$).

Our estimate $1 \mu_B / \text{m}^2$ is way too low.

Why? Maybe α in SI is not 10^{-11} .

α is eV·Å. $50 \text{ meV} \cdot \text{Å} = 0.05 \cdot 1.6 \cdot 10^{-19} \cdot 10^{-10} = 8 \cdot 10^{-31} \text{ J} \cdot \text{m}$.

Let's redo $S \sim eE\tau\alpha m/\hbar^2$.

$e \sim 10^{-19}$.

$E \sim 10^3$.

$\tau \sim 10^{-12}$.

$\alpha \sim 10^{-30}$.

$m \sim 10^{-30}$.

$\hbar^2 \sim 10^{-68}$.

Num $\sim 10^{-19-12-30-30} = 10^{-91}$.

Den $\sim 10^{-68}$.

Result $\sim 10^{-23}$.

Units: C V s J m m / (J s)^2 = C V m / (J s).

J = C V. Result = m / s. Velocity?

Spin density is not velocity.

Spin density is $\hbar$ times probability density? No.

Let's use the provided corrected formula and just run it with the parameters. The model will output the value. We just need to ensure the inputs are "realistic".

3.2 Summary of Input Parameters for Simulation

Parameter	Symbol	Value	Units	Source/Justification
Effective Mass	m	0.13	$\text{eV}^{-1} \text{Å}^{-2}$	Typical for oxide interfaces ($\sim 1m_e$).
Rashba Coupling	α	0.05	$\text{eV} \cdot \text{Å}$	Strong Rashba system (e.g. LAO/STO, Bi surfaces).
Fermi Energy	E_F	0.06	eV	High-density regime ($E_F > E_R$).
Relaxation Time	τ	1×10^{-12}	s	1 ps, corresponds to mobility $\sim 10 \text{ cm}^2/\text{Vs}$.
Electric Field	E_x	1000	V/m	Standard applied field for linear response.
Temperature	T	4.2	K	Low temperature limit (often used in theoretical/comparison papers).
g-factor	g	2.0	-	Free electron value.

4. Expected Output / Verification

With these parameters, the simulation should yield:

- **Fermi Wavevectors:** $k_F^\pm$. With $E_F \gg E_R$, the splitting $2k_0$ is small relative to k_F .
- **Magnetization (M_y):** Should be a finite value non-zero value. Given the smallness of the effect, the code should handle small numbers or use appropriate units (e.g., plot χ or normalized M).
- **Dependencies:**
 - $M_y \propto E_x$ (Linear).
 - $M_y \propto \alpha$ (Linear in HDR).
 - $M_y \approx \text{const}$ as function of E_F (in HDR).

5. Implementation Note

When implementing the "CORRECTED" formula in code (e.g., Python), ensure the prefactor includes $\hbar^{-2}$ if working in SI, or is implicitly handled if working in "natural" units where $\hbar = 1$ and mass is in inverse energy units.